SURE ProEd
CYBER SECURITY & ETHICAL HACKING INTERNSHIP

MINI PROJECT – 1: BUILD YOUR OWN CYBER SECURITY LAB

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A. Kali Linux Installation on VMware Workstation Pro 17

Step-by-step installation documentation with screenshots

Purpose and real-world application

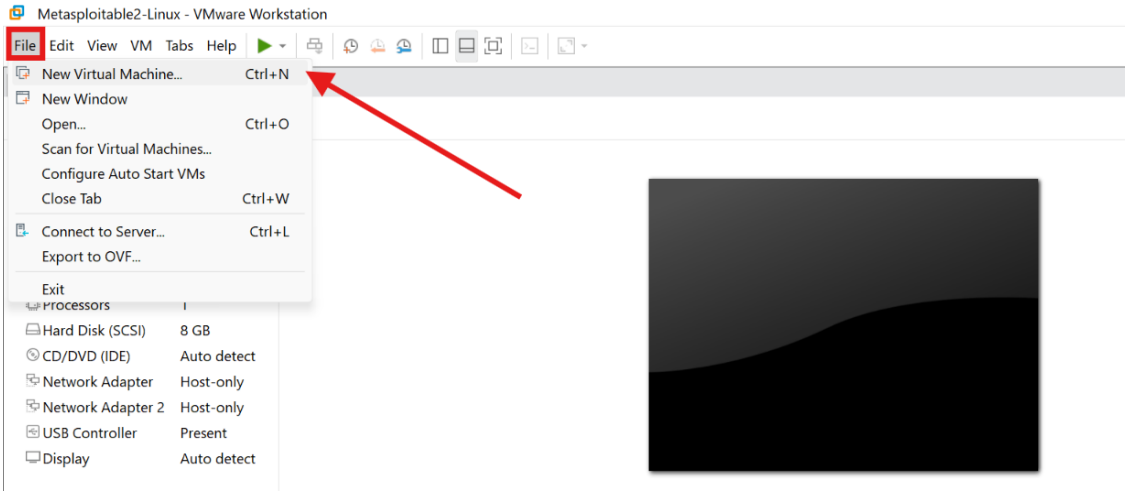
This document records how Kali Linux was installed as a virtual machine in VMware Workstation Pro 17. The VM is the attacker machine of a safe, isolated penetration-testing practice lab. A junior security analyst can scan and attack the intentionally vulnerable Metasploitable2 VM (already in the VMware library, see Figure 1) without touching any real network or system, which is how ethical-hacking skills are practised legally before real engagements.

Lab environment at a glance

Hypervisor	VMware Workstation Pro 17
Guest OS	Kali Linux (installer ISO, 2026.2 release),
VM name	kali
Resources	8 GB RAM, 1 processor × 4 cores
Disk	Split into multiple files; ext4 root + swap, GRUB on /dev/sda
Network	Adapter 1: NAT
Target VM in same library	Metasploitable2-Linux (Host-only adapters)

Part A – Creating the virtual machine in VMware Workstation Pro 17

Figure 1 – Opening the New Virtual Machine Wizard



What this screen does

The VMware Workstation window with the **File** menu open. The library already holds the Metasploitable2-Linux VM (powered off, Host-only network adapters), which is the target machine for the lab. The red arrow marks **New Virtual Machine…** (Ctrl+N).

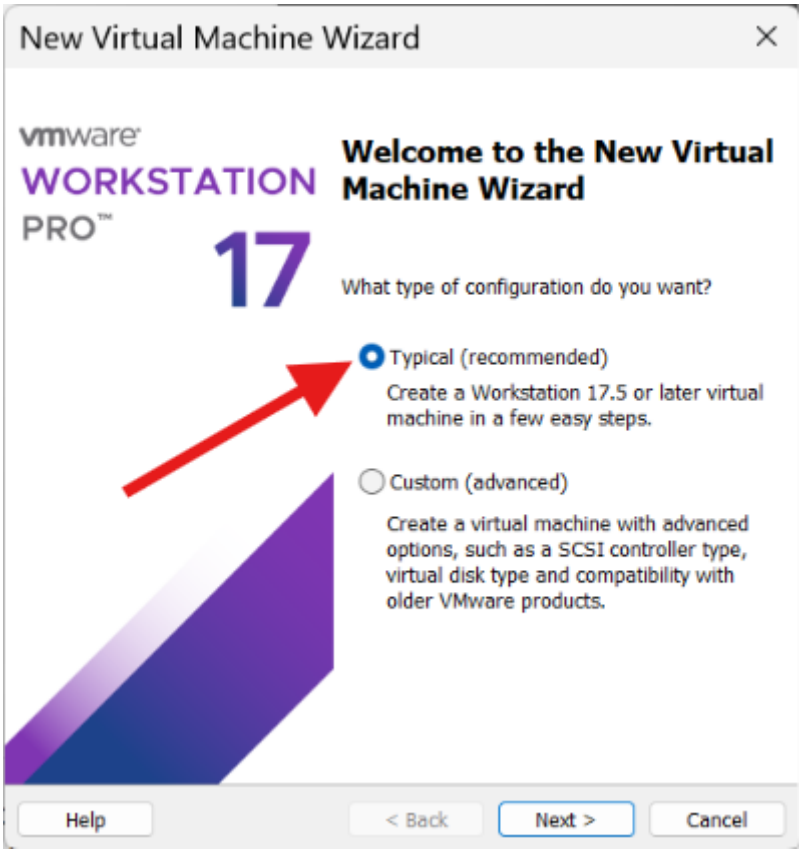
Other options on this screen

- **New Window** – opens a second Workstation window; it does not create a VM.
- **Open…** (Ctrl+O) – opens an existing VM file (.vmx) or OVF. This is how a ready-made VM such as Metasploitable2 is added.
- **Scan for Virtual Machines…** – searches a folder for existing VMs and adds them to the library.
- **Configure Auto Start VMs** – makes selected VMs start automatically with the host.
- **Close Tab** (Ctrl+W) – closes the current VM tab.
- **Connect to Server…** (Ctrl+L) – connects to a remote ESXi/vCenter/Workstation server.
- **Export to OVF…** – exports a VM into a portable OVF package.
- **Exit** – closes VMware Workstation.

Option chosen and why

> **Chosen: New Virtual Machine… (Ctrl+N).** Kali has to be installed from scratch from an ISO, so a brand-new VM is needed. Open/Scan only work for VMs that already exist, which is the case for Metasploitable2 but not for Kali.

Figure 2 – Choosing the Configuration Type (Typical)



What this screen does

The first page of the New Virtual Machine Wizard asks how much control you want over the VM's setup.

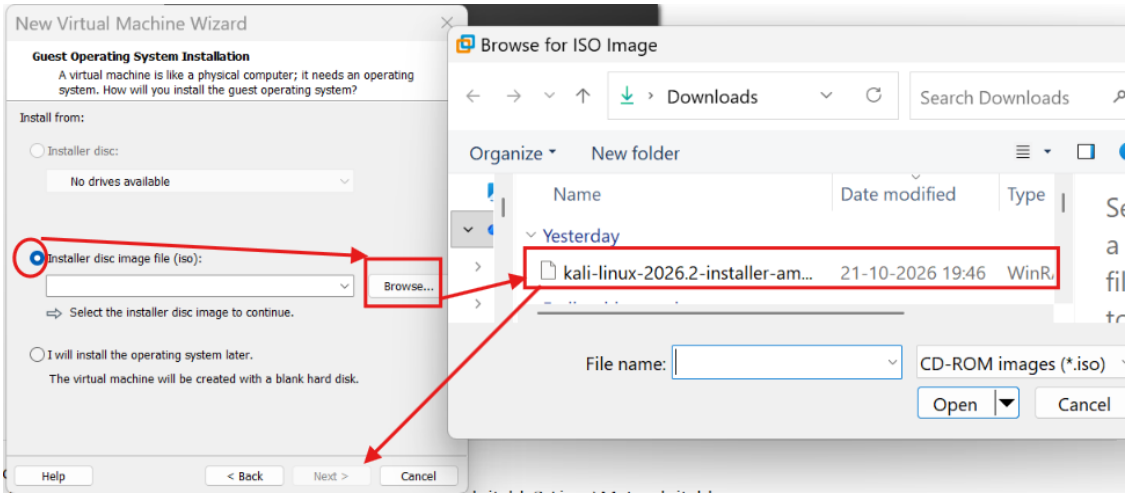
Other options on this screen

- **Custom (advanced)** – lets you pick the SCSI controller type, the virtual disk type, and the hardware compatibility level for older VMware products.

Option chosen and why

> **Chosen: Typical (recommended).** Kali has no need for a special disk controller or for compatibility with old VMware versions. Typical creates the VM in fewer steps, and the values that matter (RAM, CPU, network) can still be adjusted later through Customize Hardware (Figure 7).

Figure 3 – Selecting the Kali Linux Installer ISO



What this screen does

The Guest Operating System Installation page, with the Browse for ISO Image window on top. The file picked is the Kali installer image.

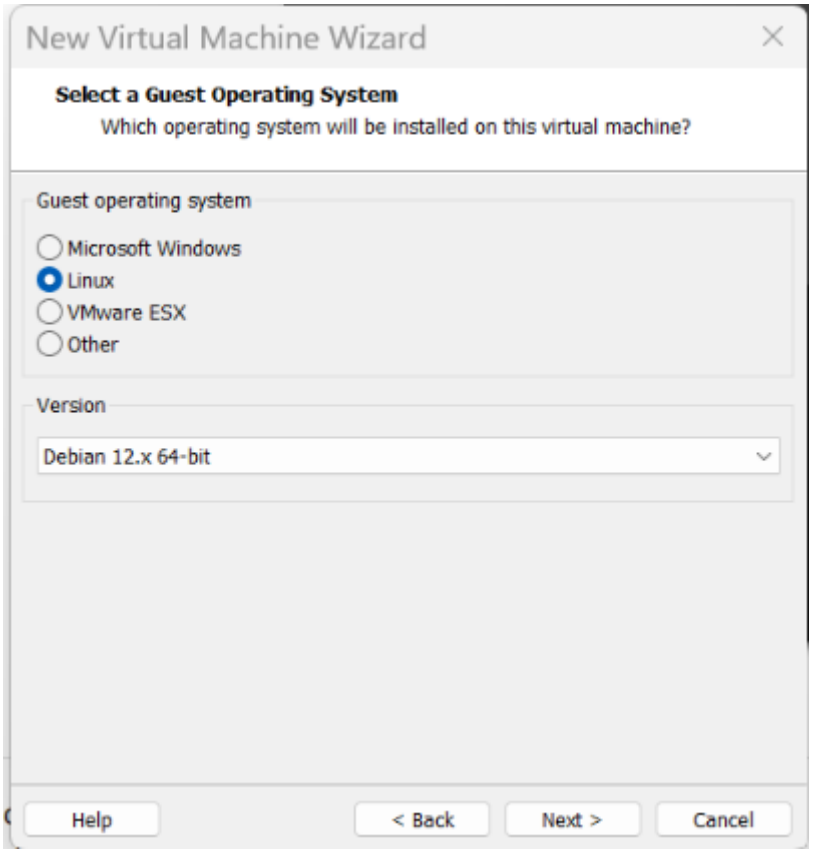
Other options on this screen

- **Installer disc** – installs from a physical CD/DVD. It is greyed out because the host reports "No drives available".
- **I will install the operating system later** – creates an empty VM with a blank hard disk; you attach installation media yourself afterwards.

Option chosen and why

> **Chosen: Installer disc image file (iso) → Browse → Kali installer ISO.** The ISO was already downloaded and there is no optical drive on the host. Selecting it here attaches it to the VM's virtual CD/DVD drive, so the VM boots straight into the Kali installer when powered on.

Figure 4 – Selecting the Guest Operating System Type



What this screen does

Tells VMware which operating system family the VM will run so it can apply suitable virtual-hardware defaults.

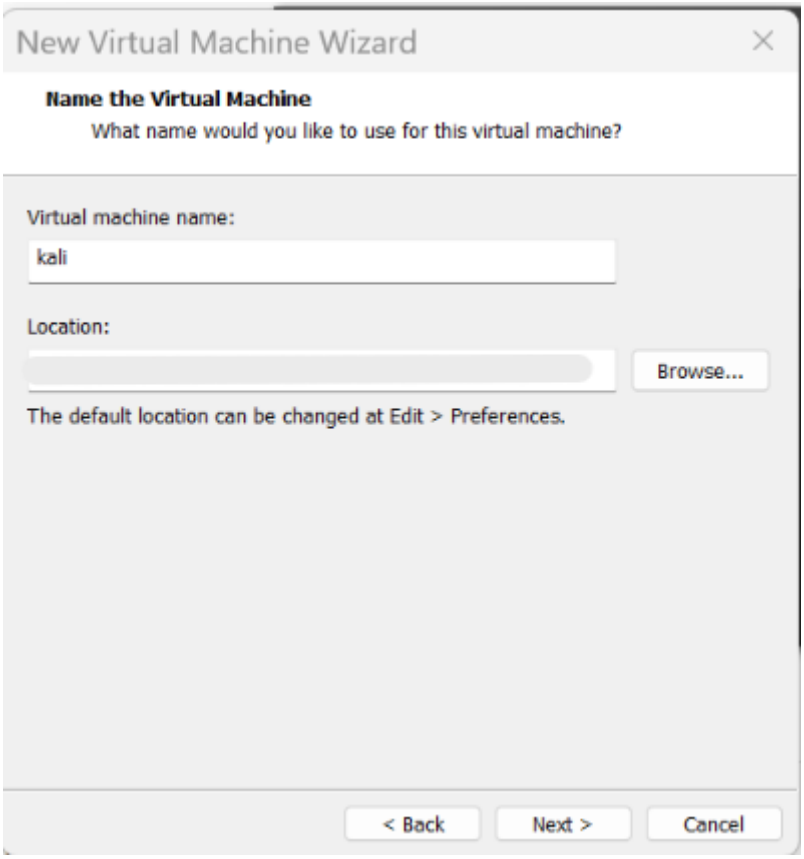
Other options on this screen

- **Microsoft Windows** – profile for Windows guests.
- **VMware ESX** – profile for nested ESXi hypervisors.
- **Other** – for operating systems that are not Windows, Linux or ESX.
- **Other Version entries** – the drop-down lists many other Linux distributions and releases.

Option chosen and why

> **Chosen: Linux → Debian 12.x 64-bit.** Kali is a Debian-based Linux distribution and VMware has no separate Kali entry. The 64-bit Debian profile gives the right defaults for a 64-bit Linux guest (including the 20 GB recommended disk size shown in Figure 6). It only affects the VM's hardware template, not the Kali version that gets installed.

Figure 5 – Naming the Virtual Machine



What this screen does

Sets the name shown in the VMware library and the folder where the VM's files are stored.

Other options on this screen

- **Other names** – any label can be used.
- **Location → Browse** – stores the VM on another drive. The default location can also be changed globally under Edit > Preferences.

Option chosen and why

> **Chosen: name "kali", default location.** A short name matches the installer hostname (Figure 8) and the tab label in the final screenshot, so it is easy to tell apart from Metasploitable2-Linux in the library. The default location was kept because the disk can grow large; another drive is only needed if the C: drive runs short of space.

Figure 6 – Specifying the Disk Capacity



What this screen does

Sets the maximum size of the virtual hard disk and how it is stored on the host. The disk starts small and grows as data is added. The wizard shows a recommended size of 20 GB for Debian 12.x 64-bit.

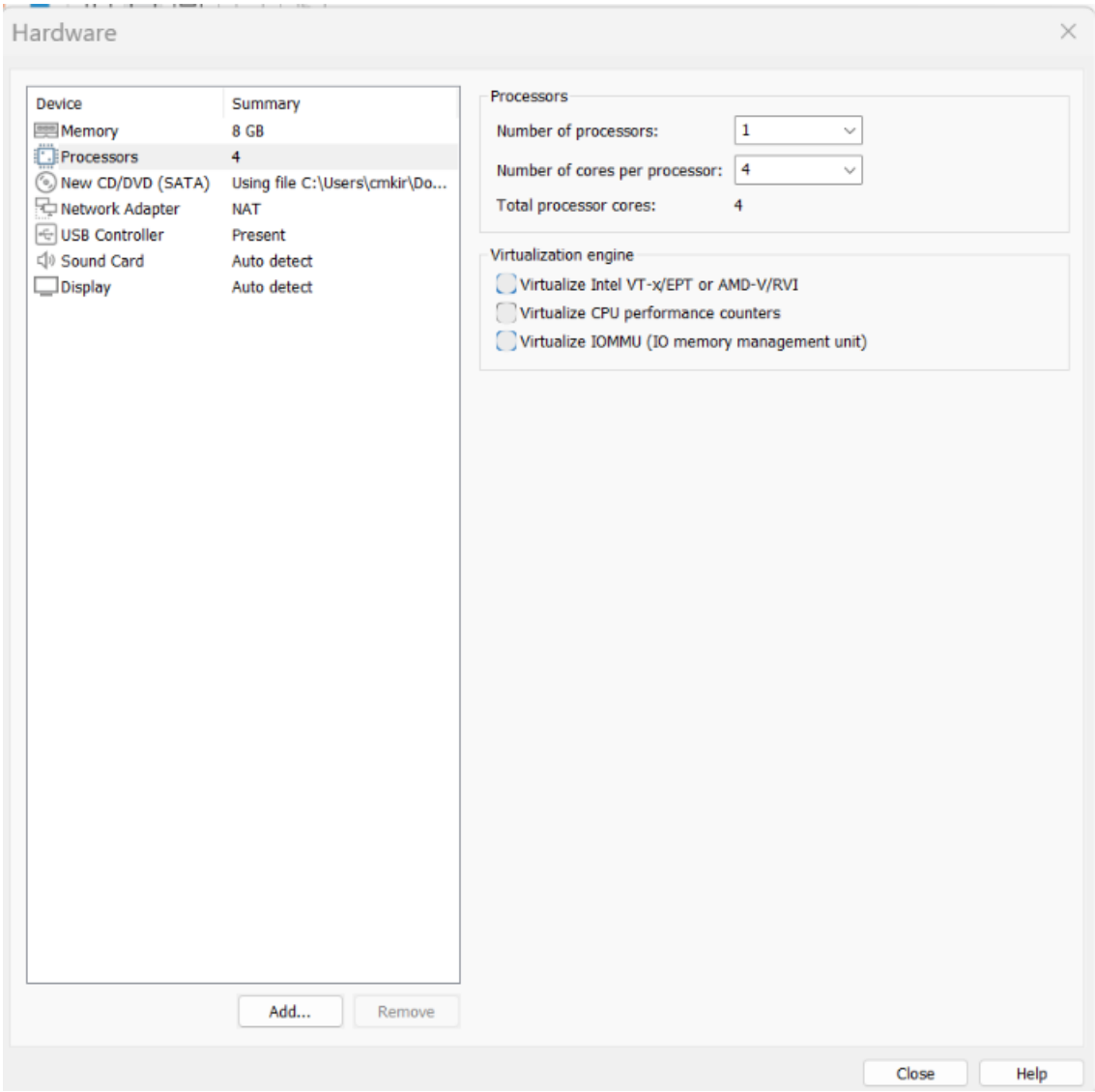
Other options on this screen

- **Store virtual disk as a single file** – slightly better performance for very large VMs. The downside is that one huge file is hard to copy or move and can fail on certain file systems (such as FAT32 drives). Best for high-performance needs on a VM that will stay on one computer.

Option chosen and why

> **Chosen: Maximum disk size 100 GB, Split virtual disk into multiple files.** Split disks are the recommended default because they are easy to copy, back up or move to another computer, and they work on FAT32 external drives. The only drawback, reduced performance, matters just for very large disks. 100 GB leaves comfortable room for Kali's tools, wordlists and capture files, well above the 20 GB recommendation.

Figure 7 – Customizing the Virtual Hardware



What this screen does

The Customize Hardware window, where RAM, CPU, virtualization features and network mode are set.

Other options on this screen

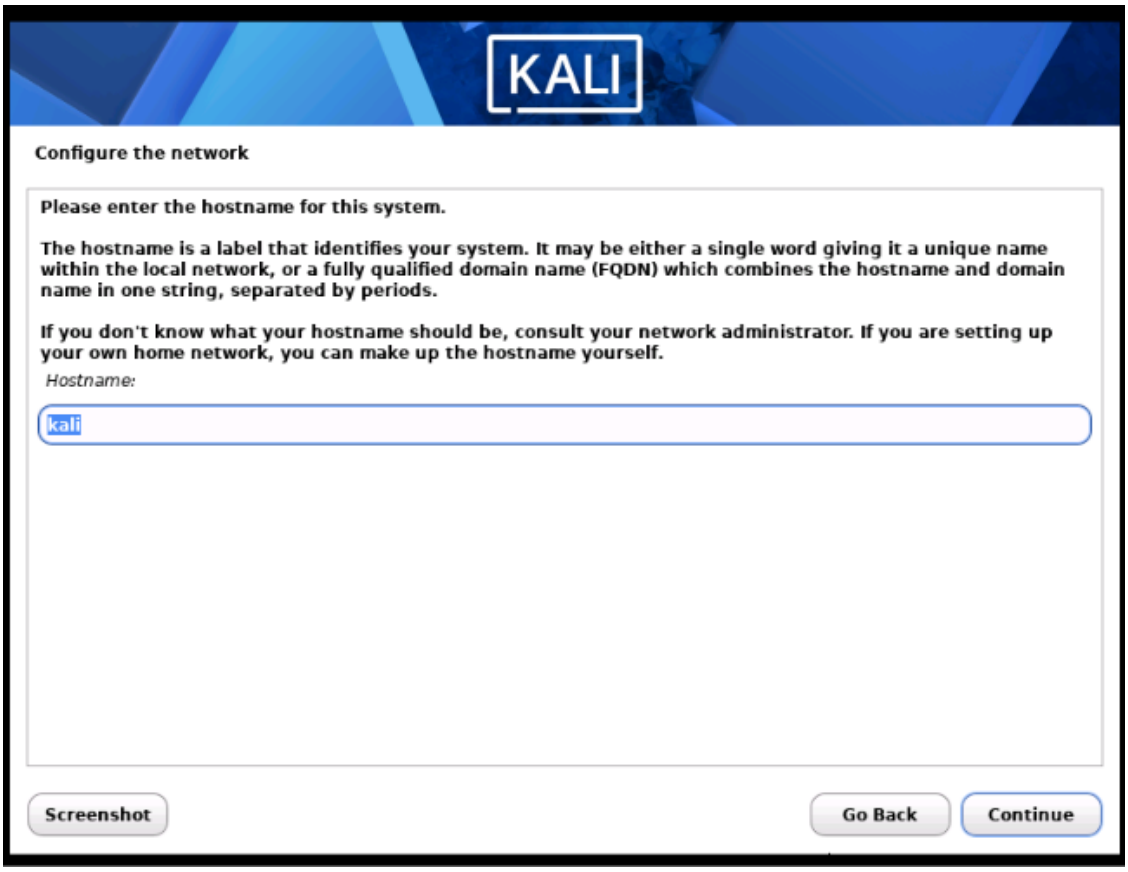
- **Memory** – any amount can be assigned; more RAM helps when several tools run together.
- **Processors** – the number of processors and cores per processor can be raised or lowered (here 1 processor × 4 cores = 4 total cores).
- **Virtualize CPU performance counters** (left unticked) – exposes CPU counters to the guest for low-level profiling tools. Not needed for normal use.
- **Network Adapter** – other modes are **Bridged** (VM appears as its own device on the physical network), **Host-only** (private network between host and VMs, no internet), **Custom** (a specific virtual network) and **LAN segment** (private network between selected VMs).
- **USB Controller, Sound Card, Display** – all left on defaults (USB present, sound and display auto-detect).
- VT-x/AMD-V gives the guest hardware-assisted virtualization for good speed
- The IOMMU option was left unticked; it is harmless and only matters for advanced features such as device pass-through.

Option chosen and why

Chosen: Memory 8 GB · Processors 1 × 4 cores . 8 GB and 4 cores let Kali's desktop run heavy tools such as Metasploit, Burp Suite and Wireshark at the same time as the Metasploitable2 target, while leaving resources for the host. NAT gives Kali internet access through the host for updates and tool installs without exposing it directly on the home or college network.

Part B – Installing Kali Linux (installer screens)

Figure 8 – Setting the Hostname



What this screen does

The first Kali installer question under "Configure the network". The hostname is the name that identifies this machine on a network.

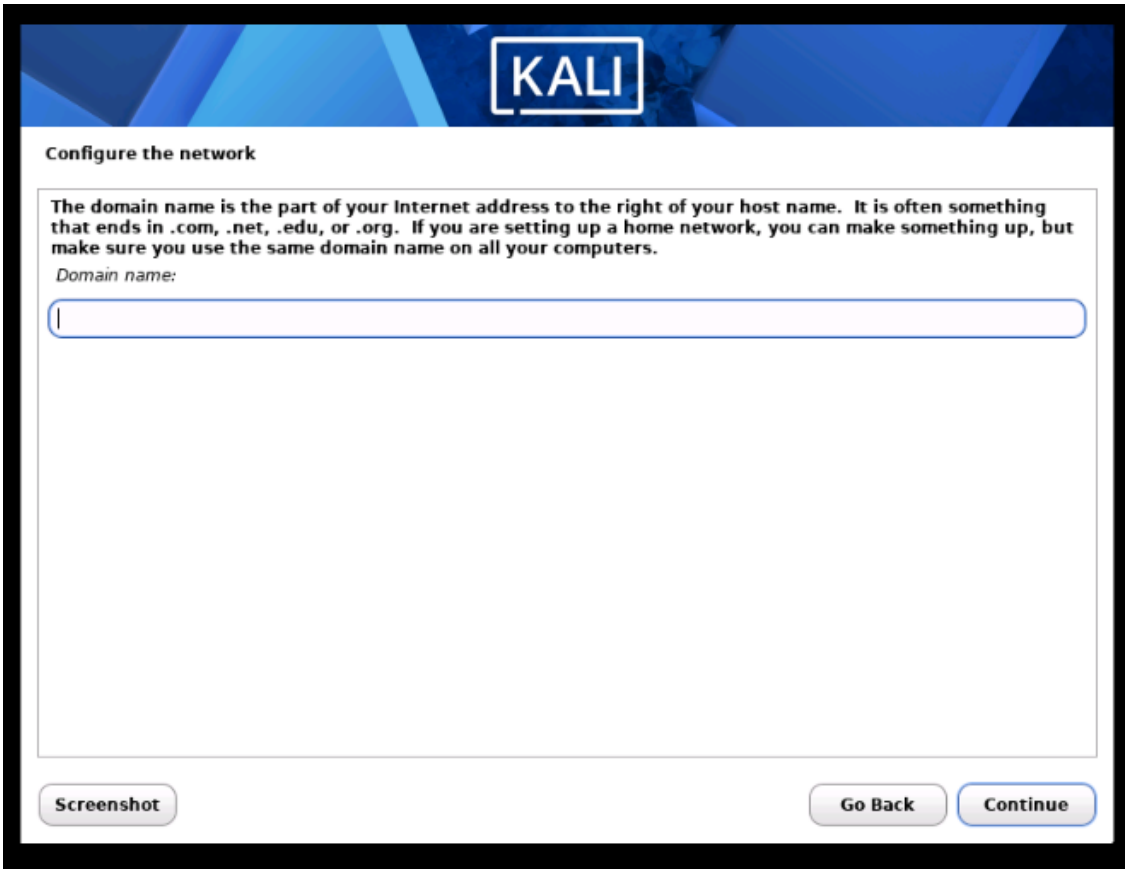
Other options on this screen

- **Any other name** – a single word or a fully qualified domain name (FQDN) can be used instead.

Option chosen and why

> **Chosen: "kali" (the default).** In the pen-test lab, the hostname is what the attacker machine is called in the target's logs and on the shell prompt (user@kali). The default is short, recognizable and matches most Kali documentation, so there was no reason to change it.

Figure 9 – Setting the Domain Name



What this screen does

Asks for the domain part of the network address (such as .com, .net or .edu).

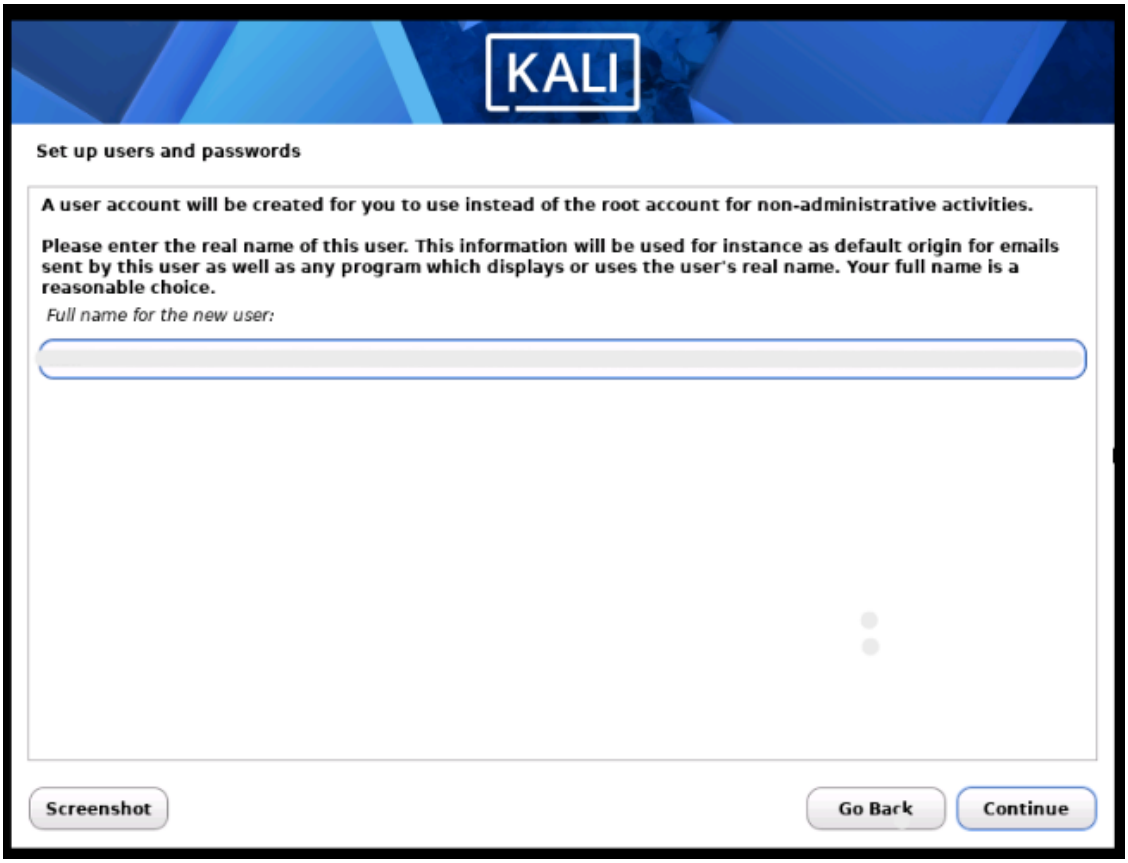
Other options on this screen

- **Entering a domain** (for example a made-up one like lab.local) – only useful if several machines should share the same domain name.

Option chosen and why

> **Chosen: left empty.** The Kali VM is a standalone machine that is not joined to any domain, and the installer allows this field to stay blank.

Figure 10 – Creating the User Account – Full Name



What this screen does

Part of "Set up users and passwords". The installer creates a normal user account instead of working as root. The full name field is blank in the capture, so the name that was entered is not visible.

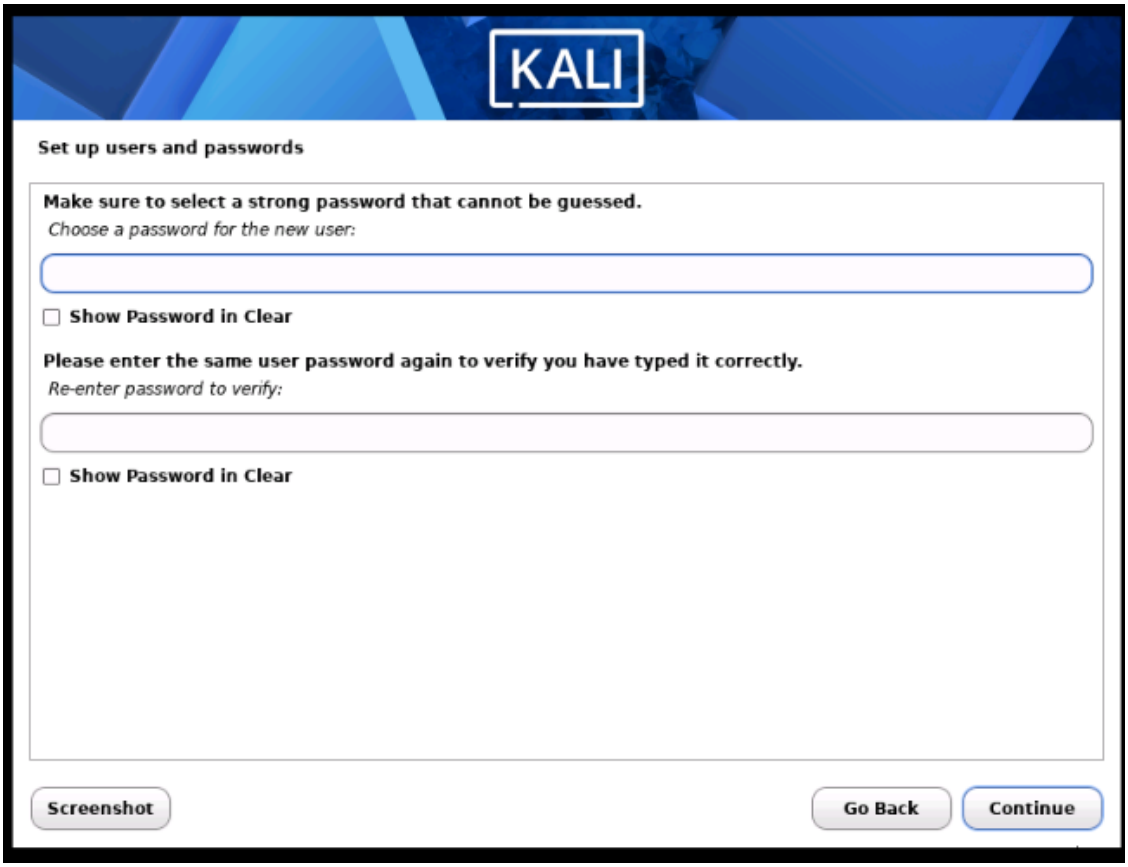
Other options on this screen

- **Using the root account for daily work** – no longer the default; the installer deliberately creates a separate user for non-administrative activities.

Option chosen and why

> **Chosen: a regular (non-root) user account.** Working as a normal user and using sudo only when needed follows the least-privilege habit used on real engagements, and it limits the damage if a tool or script misbehaves.

Figure 11 – Setting the User Password



What this screen does

Asks for the new user's password twice to avoid typos. Each box has a Show Password in Clear checkbox.

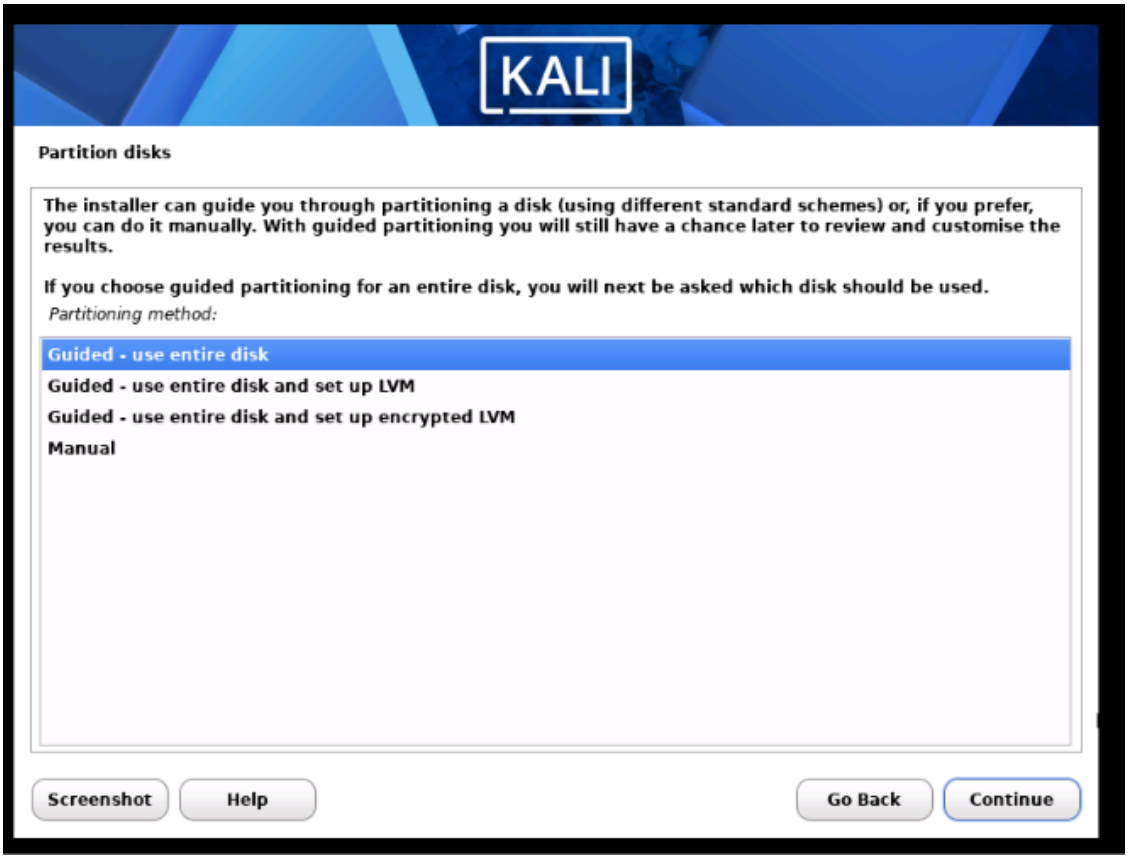
Other options on this screen

- **Show Password in Clear** (both boxes unticked) – displays the characters while typing; useful only to double-check a typo.

Option chosen and why

> **Chosen: a strong password, typed with masking on.** The VM will hold tools, captured data and test credentials, and it can be reached from the host-only lab network, so a strong password is preferable to a weak or default one. Masking prevents shoulder-surfing.

Figure 12 – Choosing the Partitioning Method



What this screen does

Asks how the installer should lay out the disk.

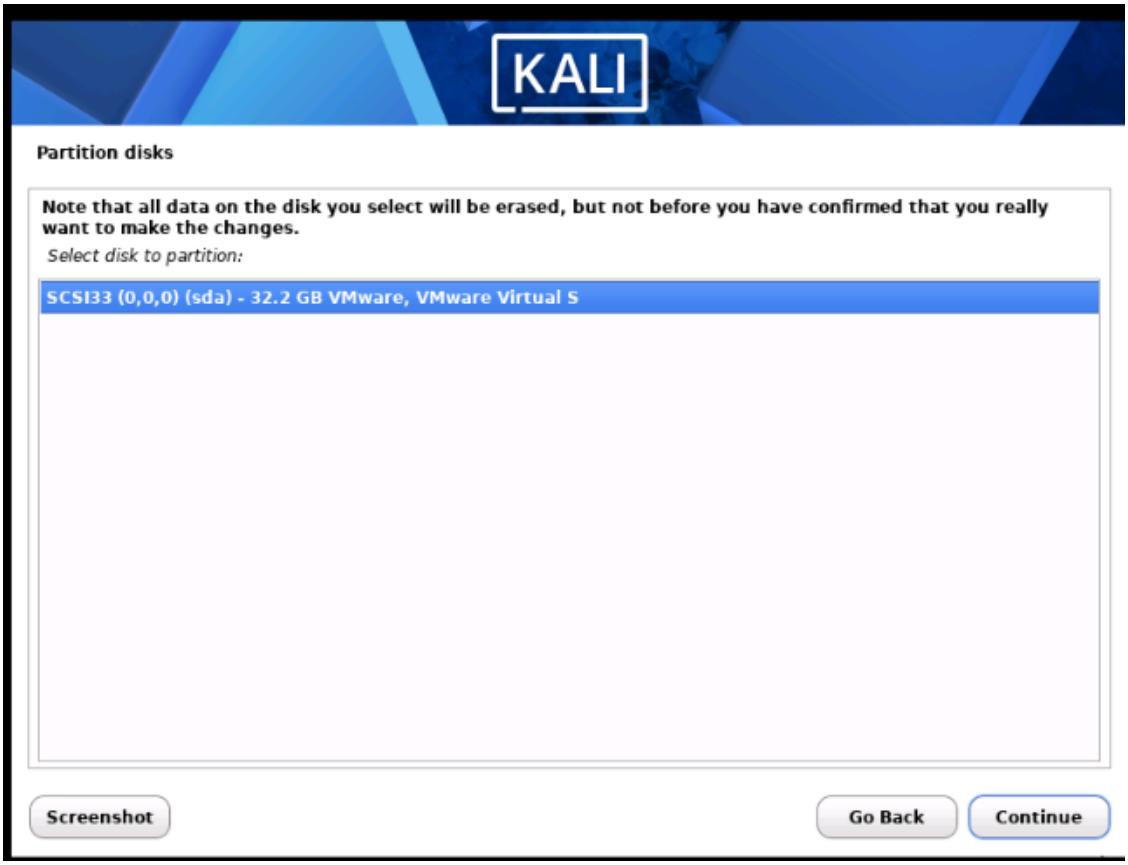
Other options on this screen

- **Guided – use entire disk and set up LVM** – uses Logical Volume Manager so volumes can be resized later.
- **Guided – use entire disk and set up encrypted LVM** – same as LVM plus full-disk encryption; a passphrase is asked at every boot.
- **Manual** – you create and size every partition yourself.

Option chosen and why

> **Chosen: Guided – use entire disk.** The disk is a virtual disk dedicated to this VM, so "entire disk" cannot touch the host's files. It is the simplest option, and LVM resizing or encryption adds nothing for a practice lab. (On a laptop VM that holds real client data, encrypted LVM would be the better choice.)

Figure 13 – Selecting the Disk to Partition



What this screen does

Lists the disks the installer can see. A warning explains that data on the chosen disk is erased, but only after the changes are confirmed (Figure 16).

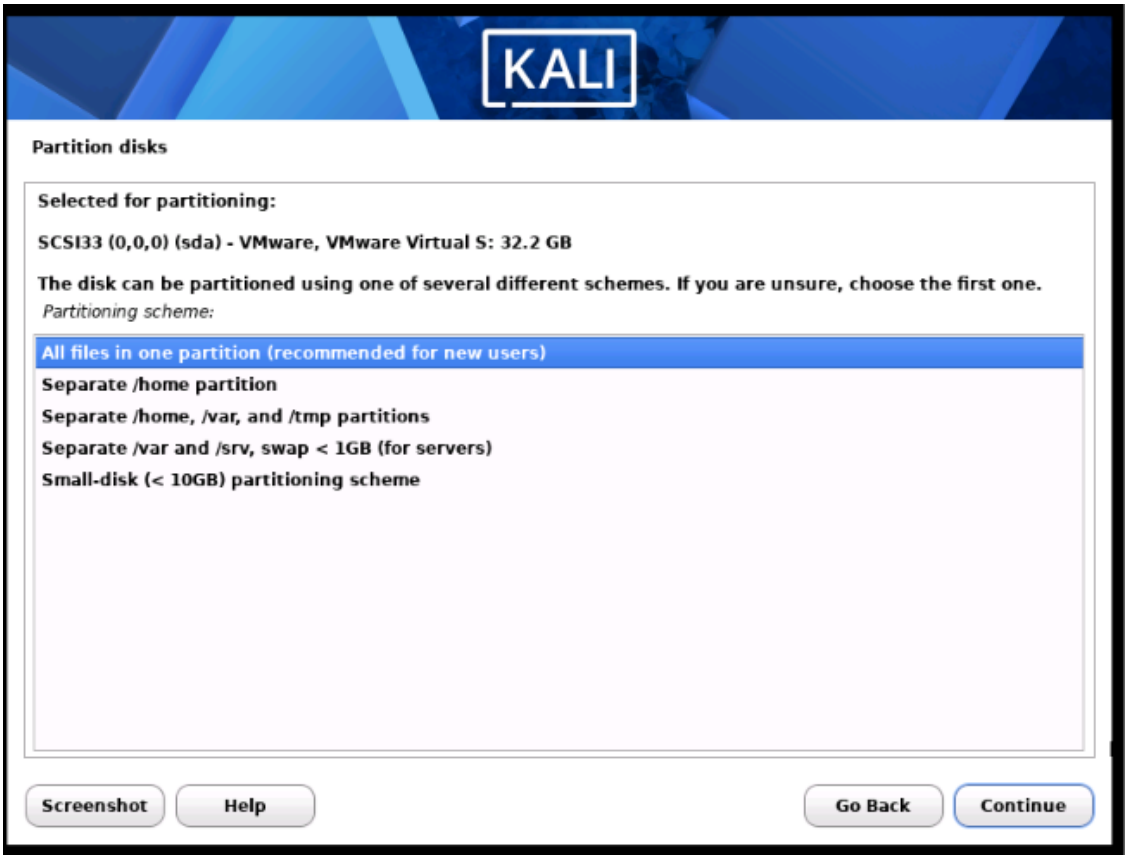
Other options on this screen

- No other options appear: the only disk listed is **SCSI33 (0,0,0) (sda) – 32.2 GB VMware, VMware Virtual S**.

Option chosen and why

> **Chosen: SCSI33 (0,0,0) (sda) – 32.2 GB.** It is the only disk, and it is the VM's own virtual disk, so nothing on the host is at risk. Note that the installer shows 32.2 GB, whereas Figure 6 showed a 100 GB maximum (see the notes at the end).

Figure 14 – Choosing the Partitioning Scheme



What this screen does

Asks how the chosen disk should be divided into partitions.

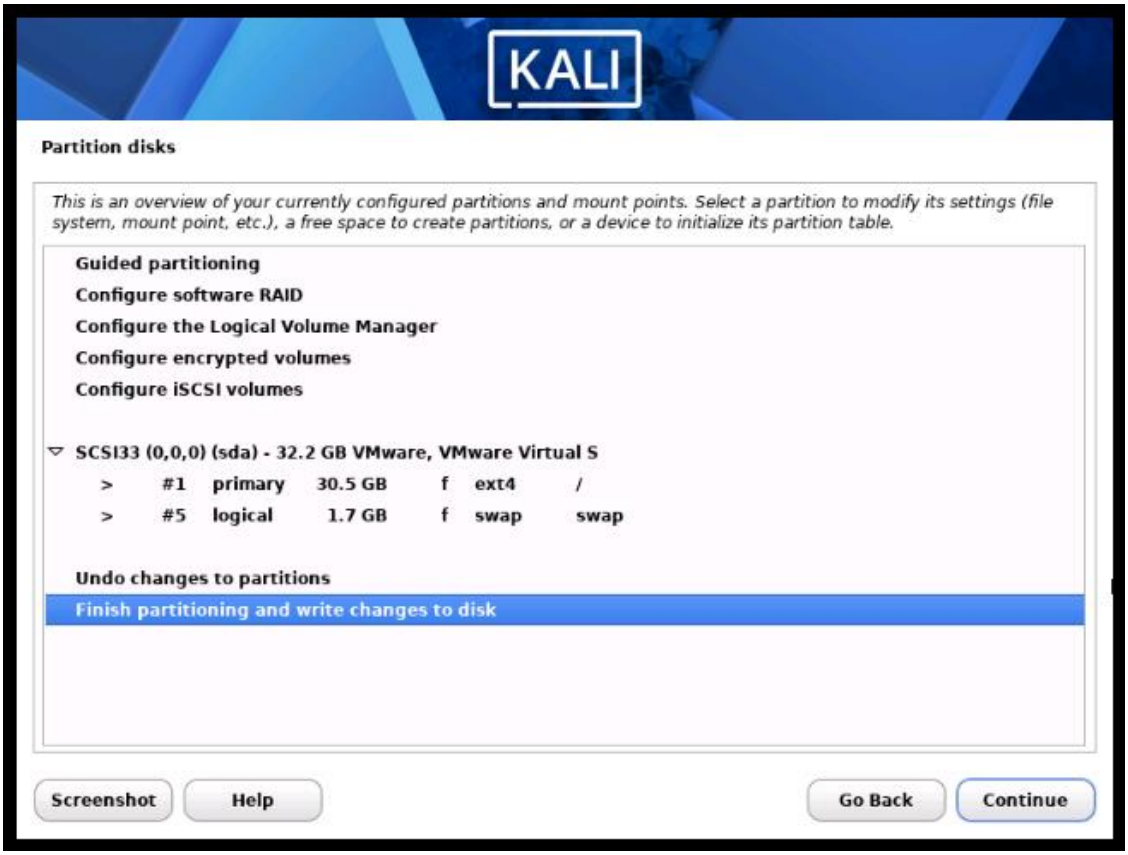
Other options on this screen

- **Separate /home partition** – keeps user files apart from the system.
- **Separate /home, /var, and /tmp partitions** – further isolates logs and temporary files.
- **Separate /var and /srv, swap < 1GB (for servers)** – layout aimed at server roles.
- **Small-disk (< 10GB) partitioning scheme** – compact layout for very small disks.

Option chosen and why

> **Chosen: All files in one partition (recommended for new users).** One partition is the simplest layout and avoids one area (such as /var or /home) filling up while others have free space, which can happen when many security tools are installed. Separate partitions mainly benefit multi-user servers.

Figure 15 – Reviewing the Partition Layout



What this screen does

A summary of the layout the installer created: **#1 primary 30.5 GB ext4 mounted on /** and **#5 logical 1.7 GB swap**.

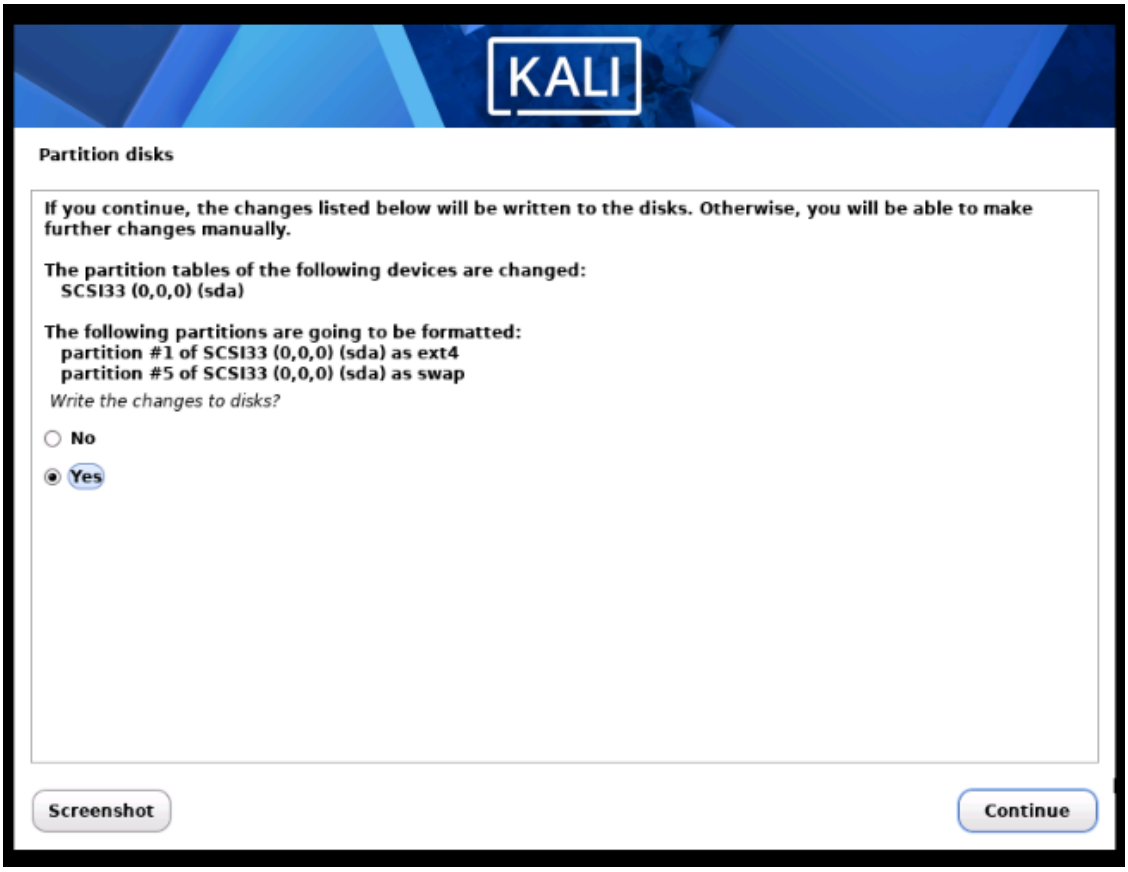
Other options on this screen

- **Guided partitioning** – re-run the guided wizard.
- **Configure software RAID / Logical Volume Manager / encrypted volumes / iSCSI volumes** – advanced storage setups.
- **Undo changes to partitions** – discards the proposed layout.
- **Selecting an individual partition** – to change its size, file system or mount point.

Option chosen and why

> **Chosen: Finish partitioning and write changes to disk.** The proposed layout (ext4 root plus swap) is exactly what a single-purpose lab VM needs, so no changes were required.

Figure 16 – Confirming the Disk Changes



What this screen does

The last chance to stop before the disk is changed. It lists that the partition table of sda will change and that partition #1 (ext4) and #5 (swap) will be formatted.

Other options on this screen

- **No** – returns to the partitioning menu for further manual changes.

Option chosen and why

> **Chosen: Yes.** The layout had been reviewed in Figure 15 and the disk is a brand-new empty virtual disk, so there is no data to lose.

Figure 17 – Installing the Base System



What this screen does

A progress bar while the installer copies and unpacks Kali's base packages ("Unpacking iproute2…").

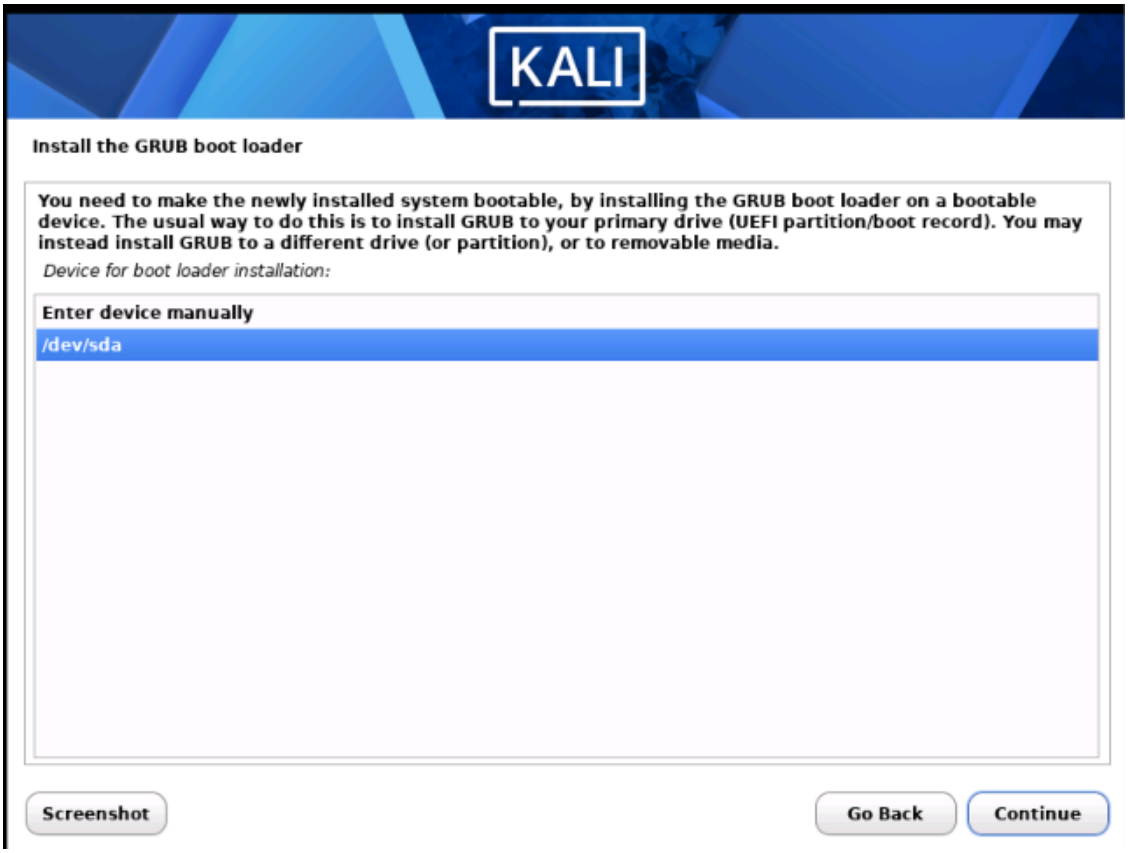
Other options on this screen

This screen has no selectable options.

Option chosen and why

> **Chosen: nothing to select; wait for it to finish.** This is an automatic step. Interrupting it would leave a half-installed system.

Figure 18 – Installing the GRUB Boot Loader



What this screen does

GRUB is the program that starts Kali when the VM powers on. The installer asks where to put it.

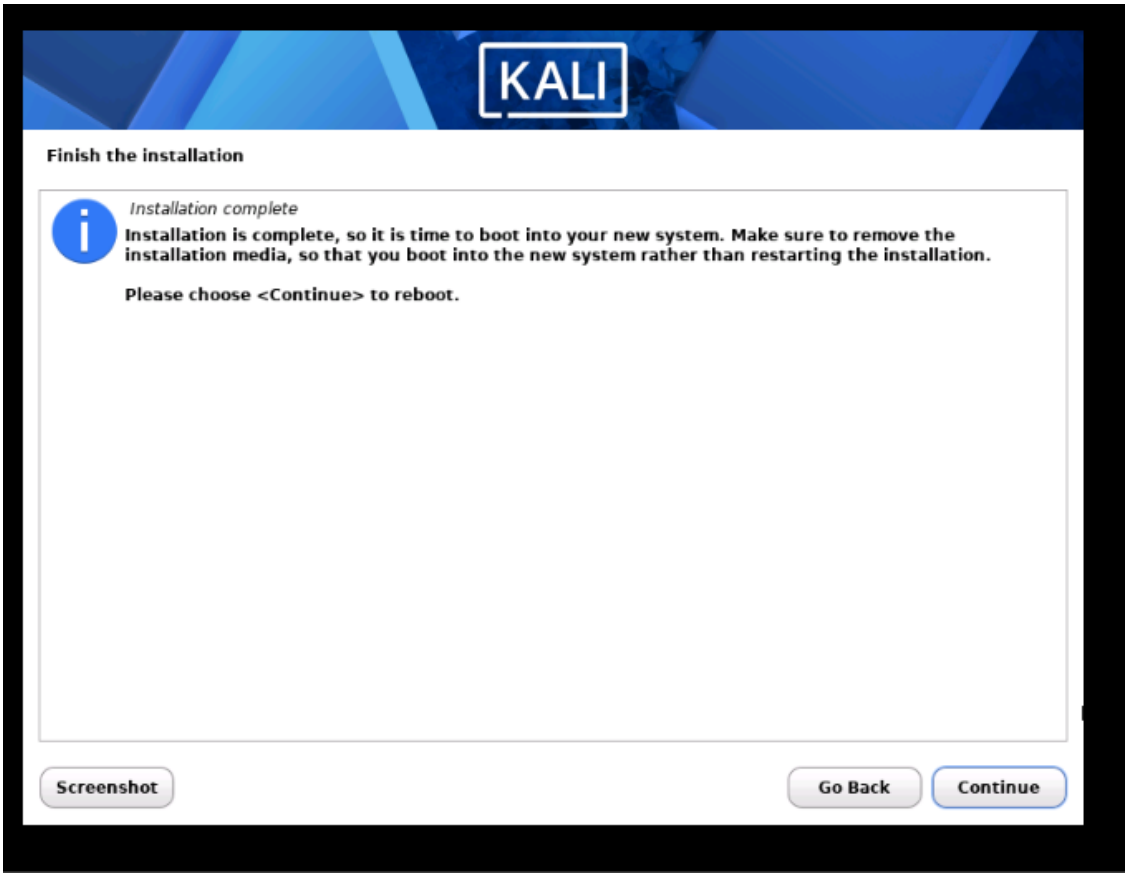
Other options on this screen

- **Enter device manually** – lets you type another device, partition or removable media for GRUB.

Option chosen and why

> **Chosen: /dev/sda.** It is the VM's only disk. Installing GRUB there makes the VM boot from its own disk without the ISO.

Figure 19 – Installation Complete



What this screen does

The installer reports that installation is finished and asks to remove the installation media before rebooting.

Other options on this screen

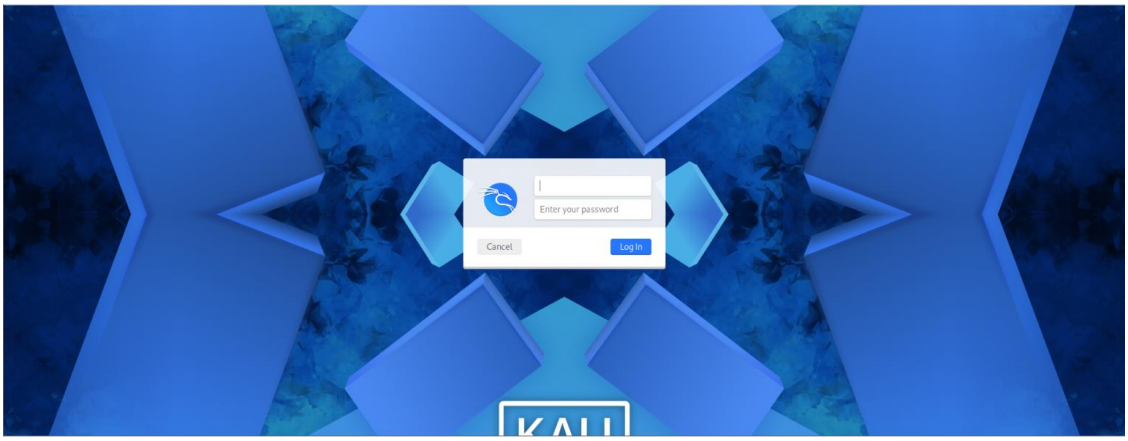
- **Go Back** – returns to the previous installer step (not needed once installation has finished).

Option chosen and why

> **Chosen: Continue (reboot).** This restarts the VM into the newly installed Kali. If the VM ever boots back into the installer, disconnect the ISO from the virtual CD/DVD drive in VM Settings.

Part C – First boot

Figure 20 – Kali Login Screen



What this screen does

The graphical login prompt shown after the first reboot, with a username box, a password box and two buttons.

Other options on this screen

- **Cancel** – dismisses the login dialog without signing in.

Option chosen and why

> **Chosen: Log In with the account created in Figures 10–11.** Logging in as the regular user (not root) keeps to the least-privilege approach used in real security work.

Figure 21 – Kali Linux Desktop (Installation Verified)



What this screen does

The Kali desktop (Xfce) with Home, File System and Trash icons, a top panel with four workspaces, and the clock showing 20:03. The VMware tabs at the top show Home, Metasploitable2-Linux and kali, so both the attacker and the target are now in the same Workstation window.

Other options on this screen

This screen has no selectable options.

Option chosen and why

> **Result: Kali Linux installed and booting successfully.** This confirms the setup is complete and the lab's attacker machine is ready.

Summary of choices

Fig.	Screenshot name	Choice made
1	Opening the New Virtual Machine Wizard	New Virtual Machine… (Ctrl+N)
2	Choosing the Configuration Type	Typical
3	Selecting the Kali Linux Installer ISO	Installer disc image file (iso)
4	Selecting the Guest Operating System Type	Linux, Debian 12.x 64-bit
5	Naming the Virtual Machine	“kali”, default location
6	Specifying the Disk Capacity	100 GB, split into multiple files
7	Customizing the Virtual Hardware	8 GB RAM, 4 cores, VT-x, IOMMU, NAT
8	Setting the Hostname	kali
9	Setting the Domain Name	Left empty
10	Creating the User Account – Full Name	Regular (non-root) user
11	Setting the User Password	Strong password, masked
12	Choosing the Partitioning Method	Guided – use entire disk
13	Selecting the Disk to Partition	sda (32.2 GB)
14	Choosing the Partitioning Scheme	All files in one partition
15	Reviewing the Partition Layout	Finish partitioning and write changes
16	Confirming the Disk Changes	Yes
17	Installing the Base System	Wait (automatic)
18	Installing the GRUB Boot Loader	/dev/sda
19	Installation Complete	Continue (reboot)
20	Kali Login Screen	Log In as the created user
21	Kali Linux Desktop (Installation Verified)	Installation verified

Notes and recommended follow-ups

- **Disk size mismatch:** Figure 6 shows a 100 GB maximum, but the installer in Figures 13–15 sees a 32.2 GB disk. Check VM Settings > Hard Disk to confirm the actual size, and correct this document if the VM was created with a different value.
- **Screens not captured:** language, location, keyboard layout, username, time zone, software selection and mirror/proxy prompts do not appear in the screenshots. Add them if a fully complete record is needed.
- **Reaching the target:** Metasploitable2 uses Host-only adapters, while Kali has only a NAT adapter. For Kali to attack Metasploitable2, add a second Kali network adapter set to the same Host-only network.
- **First actions after login:** update the system (sudo apt update, then sudo apt full-upgrade), and take a VMware snapshot named “Clean install” so the lab can be reset after each practice session.
- **Reasons given:** the “why” statements are written from the screenshots and typical lab practice. Edit any that do not match your own reasoning.

B. Linux Basic Commands in Kali Linux on VMware Workstation Pro 17

Step-by-step installation documentation with screenshots

Figure 1 – View Command History

The history command displays a list of previously executed commands in the terminal.

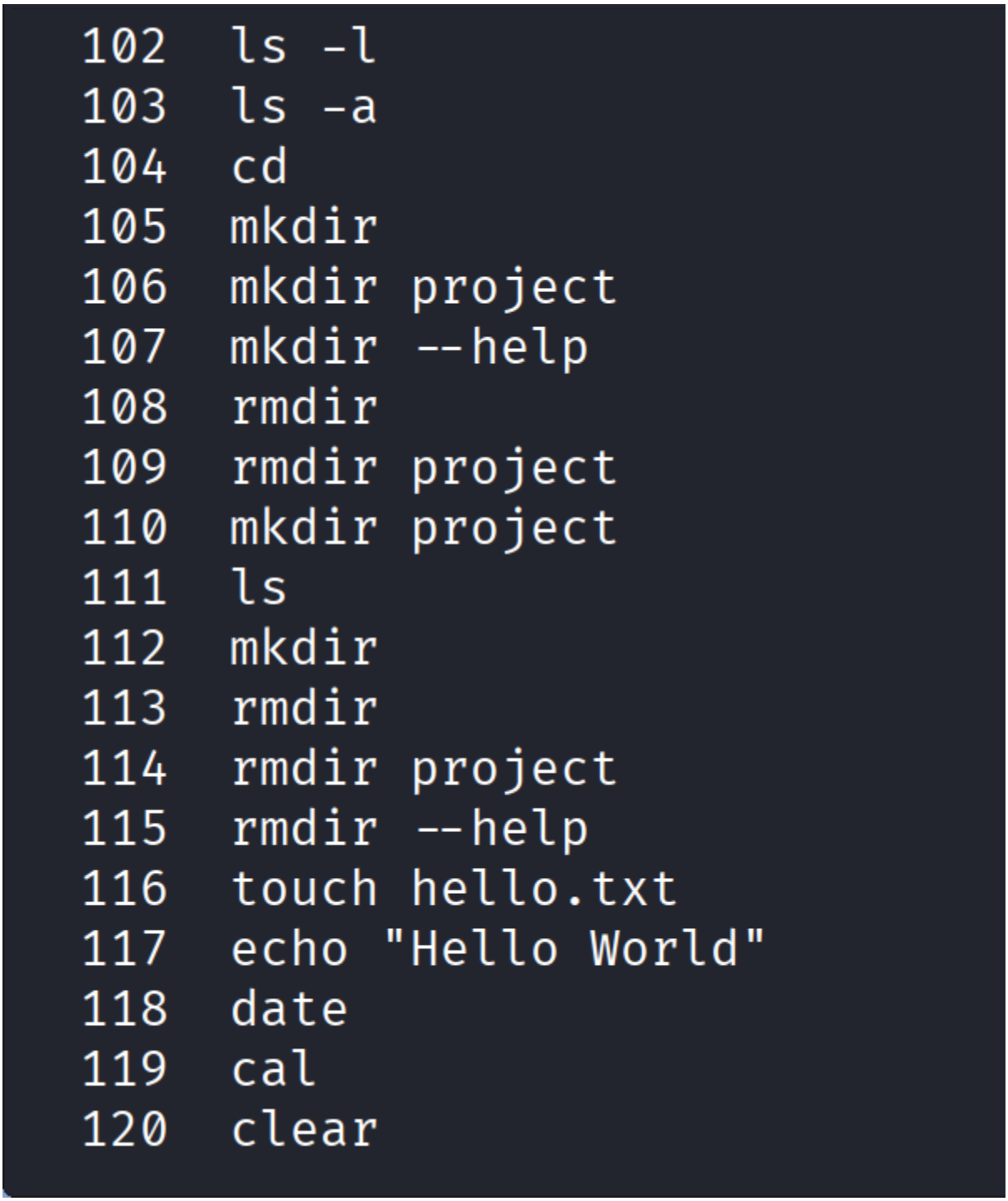


Figure 2 – List Files and Directories

The ls command displays the names of files and directories in the current directory.

```
└─$ ls
Desktop    Downloads  Pictures   Templates
Documents  Music      Public     Videos
```

Figure 3 – Detailed File Listing

> The ls -l command displays files and directories in long format, including permissions, ownership, file size, and modification date.

```
└─$ ls -l
total 32
drwxr-xr-x 2 [redacted] Desktop
drwxr-xr-x 2 [redacted] Documents
drwxr-xr-x 2 [redacted] Downloads
drwxr-xr-x 2 [redacted] Music
drwxr-xr-x 2 [redacted] Pictures
drwxr-xr-x 2 [redacted] Public
drwxr-xr-x 2 [redacted] Templates
drwxr-xr-x 2 [redacted] Videos
```

Figure 4 – Create a File

> The touch command creates an empty file. If the file already exists, it updates its timestamp.

```
└─$ touch hello.txt
```

Figure 5 – Display Text

> The echo command displays text on the terminal. It can also be used to write text into a file.

```
└─$ echo "Hello World"
Hello World
```

Figure 6 – Print Working Directory

> The pwd command displays the full path of the current working directory.

```
└─$ pwd
/home/[redacted]
```

Figure 7 – Create a directory

> The mkdir command creates a new directory (folder).

```
└─$ mkdir project

┌──[redacted]
└─$ mkdir --help
Usage: mkdir [OPTION]... DIRECTORY...
Create the DIRECTORY(ies), if they do not already exist.

Mandatory arguments to long options are mandatory for short options too.
  -m, --mode=MODE   set file mode (as in chmod), not a=rwx - umask
  -p, --parents     no error if existing, make parent directories as need
ed,
                    with their file modes unaffected by any -m option
  -v, --verbose     print a message for each created directory
  -Z                   set SELinux security context of each created direc
tory
                         to the default type
      --context[=CTX]  like -Z, or if CTX is specified then set the SELin
ux
                         or SMACK security context to CTX
      --help        display this help and exit
      --version     output version information and exit

GNU coreutils online help: <https://www.gnu.org/software/coreutils/>
Full documentation <https://www.gnu.org/software/coreutils/mkdir>
or available locally via: info '(coreutils) mkdir invocation'
```

Figure 8 – Remove an Empty Directory

> **The rmdir command removes a directory only if it is empty.**

```
└─$ rmdir
rmdir: missing operand
Try 'rmdir --help' for more information.

┌──[redacted]
└─$ rmdir project

┌──[redacted]
└─$ rmdir --help
Usage: rmdir [OPTION]... DIRECTORY...
Remove the DIRECTORY(ies), if they are empty.

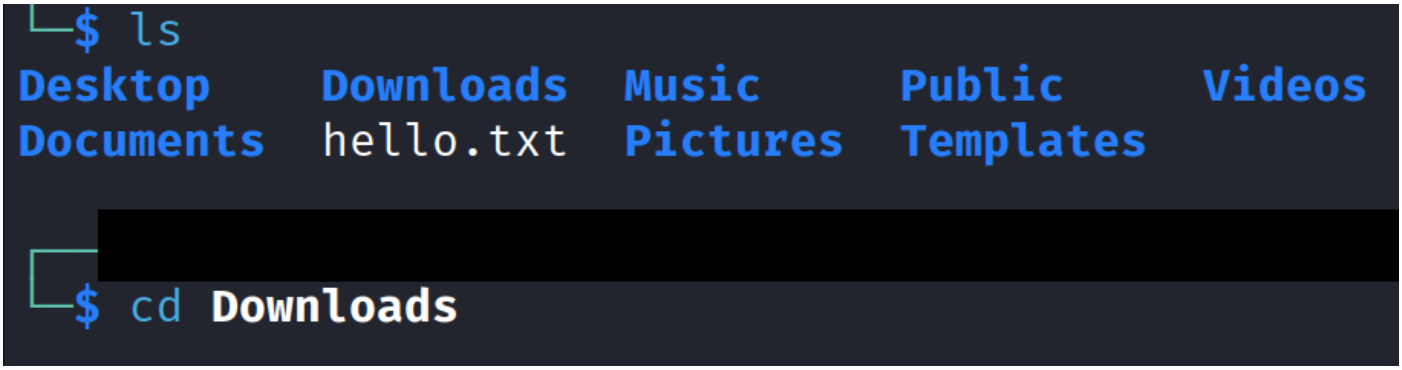
      --ignore-fail-on-non-empty
                    ignore each failure to remove a non-empty directory
  -p, --parents     remove DIRECTORY and its ancestors;
                    e.g., 'rmdir -p a/b' is similar to 'rmdir a/b a'

  -v, --verbose     output a diagnostic for every directory processed
      --help        display this help and exit
      --version     output version information and exit

GNU coreutils online help: <https://www.gnu.org/software/coreutils/>
Full documentation <https://www.gnu.org/software/coreutils/rmdir>
or available locally via: info '(coreutils) rmdir invocation'
```

Figure 9 – Change Directory

> **The cd command is used to move from one directory to another.**


$ ls
Desktop Downloads Music Public Videos
Documents hello.txt Pictures Templates
$ cd Downloads